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System-scale design and analysis of desalination systems for meeting water needs of data centers

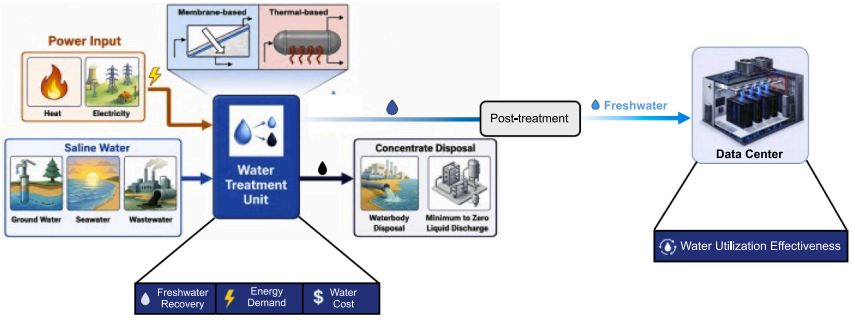
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HIGHLIGHTS

- Techno-economic assessment of desalination systems for data centre use.
- Novel SWUI metric introduced for data center water-stress impact assessment.
- RO and MVC analyzed for brackish to hypersaline PW feeds till MLD-ZLD limits.
- MLD and ZLD based treatment increase energy and cost by >12%.
- Evaporative cooling causes severe water stress with SWUI > 7.

GRAPHICAL ABSTRACT



ARTICLE INFO

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Produced water
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ABSTRACT

Buildout of data centers for high-performance computing and AI has intensified energy demands, which commensurately increases freshwater requirements for thermal management. This study presents technical and techno-economic analyses of desalination systems to produce freshwater (Total dissolved solids <500 ppm) for data centers. Two-stage reverse osmosis (RO), Mechanical Vapor Compression (MVC), and hybrid configurations are evaluated for treatment of brackish (5000 ppm), seawater (35,000 ppm), and produced water (≥50,000 ppm) feeds. Brine management is via conventional disposal or minimum to zero liquid discharge (MLD, ZLD). Freshwater recovery (FR), energy consumption (Winp), and water cost (WC) are quantified for various technologies. Results show that transitioning from conventional disposal to MLD and ZLD increased FR by 4–220% across feed salinities, while Winp and WC rise >12% and >15%, respectively. Results show FR = 736 L/1000 kg-feed, Winp = 9.4 kWh/m^3-permeate and WC = 3.2 $/m^3-permeate for seawater hybrid RO-MVC-based treatment with MLD disposal. For MVC-based treatment (with MLD) of 50,000 ppm PW, we obtain FR = 753 L/1000 kg-feed, Winp = 36 kWh/m^3-permeate and WC = 10.4 $/m^3-permeate. A novel performance metric, Saline Water Utilization Intensity (SWUI) is introduced, which quantifies data center desalination impact on local water stress. Across representative conditions, SWUI increased from 2023–2030, spanning mild to substantial stress impacts. Higher recovery through MLD reduces SWUI by 40% in water-stressed settings, enabling effective water-resource utilization despite energy and cost tradeoffs. Overall, this study establishes a detailed modeling framework for sustainability-driven assessment of desalination-integrated data center infrastructure.

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Nomenclature

Acronyms

AI	Artificial intelligence
BW	Brackish water
CEPCI	Chemical Engineering Plant Cost Index
ERD	Energy Recovery Device
IT	Information technology
MLD	Minimum liquid discharge
MVC	Mechanical vapor compression
PX	Pressure Exchanger
PW	Produced water
RO	Reverse osmosis
SW	Seawater
SIE	Solar-driven Interfacial Evaporation
ZLD	Zero-liquid discharge

Symbols

A	Membrane water permeability coefficient ($m^3\ m^{-2}\ s^{-1}\ Pa^{-1}$)
A_e	Evaporator–condenser heat exchanger area (m^2)
A_{hx}	Heat exchanger area (m^2)
A_{ph}	Feed preheater heat exchanger area (m^2)
A_m	Membrane area (m^2)
ACE	Annual Capital Expenditure ($)
AOE	Annual Operational Expenditure ($)
AWP	Annual water production ($m^3\ yr^{-1}$)
B	Salt Permeability Coefficient (m^3/m^2-s)
c	Concentration (ppm)
C_{pv}	Specific heat capacity at constant pressure of water vapor
CF	Amortized Capitalization Factor
d_h	Hydraulic diameter (m)
FR	Freshwater recovery (L/1000 kg-feed)
h_{fg}	Net change in enthalpy from saturated liquid to saturated vapor (kJ/kg)
$J_{w1,2}$	Water flux at RO stage inlet and exit ($m^3\ m^{-2}\ s^{-1}$)
$LMTD_{HX}$	Logarithmic mean temperature difference across heat exchanger (K)
MRR	Membrane Rejection Ratio (–)
PUE	Power Utilization Effectiveness (–)
PIF	Practical Investment Factor (–)
PR	Pressure Ratio
ΔP	Transmembrane pressure difference (bar)
ΔP_{max}	Maximum allowable transmembrane pressure (bar)
$\dot{q}$	Volumetric flow rate ($m^3\ hr^{-1}$)
$\dot{Q}$	Rate of heat transfer (W)
RP	Permeate recovery (–)
$SWUE$	Saline Water Utilization Effectiveness ($L\ kWh^{-1}$)
$SWUI$	Saline Water Utilization Intensity ($L\ kWh^{-1}$)
T	Temperature (K)
ΔT_{ttd}	Terminal temperature difference (K)
U	Heat transfer coefficient ($kW\ m^{-2}\ K^{-1}$)
WC	Water treatment cost ($\$\ m^{-3}$)
W_{inp}	Specific work input ($kWh\ m^{-3}$)
W_{least}	Least work of separation ($kWh\ m^{-3}$)
WS	Water Stress (–)
WUE	Water Utilization Effectiveness ($L\ kWh^{-1}$)

Greek Symbols

α	Work distribution parameter in two-stage RO (–)
η_P	Pump efficiency (–)
η_{PX}	Pressure exchanger efficiency (–)
η_{comp}	Compressor isentropic efficiency (–)
η_{II}	Second-law efficiency (%)
ρ	Density ($kg\ m^{-3}$)
γ	Heat capacity ratio (–)
π	Osmotic pressure (Pa)

Subscripts

b	Brine
$chem$	Chemicals
eci	Evaporator-Condenser inlet
f	Feed
fi	Feed inlet
fe	Feed exit
p	Permeate
ph	Preheater
c	Concentrate
co	Compressor Exit
e	Exit
$elec$	Electricity
$evap$	Evaporator
$satco$	Saturation Condition at Compressor Exit
avg	Average
i	Inlet
$InPre$	Pretreatment
o	Outlet
$1,2$	Stage number
RO	Reverse osmosis stage
MVC	Mechanical vapor compression stage
pre	Pretreatment
$post$	Post-treatment
$pump$	Pump
PX	Pressure Exchanger
mem	Membrane
mix	Preheated feed mixer
ml	Maintenance and Labor
sat	Saturation condition
$solv$	Solvent water
ZLD	Zero-liquid discharge

Superscript

$'$	Intermediate Stage Condition

1. Introduction

Modern data centers underpin the digital economy, enabling cloud computing, artificial intelligence, and real-time analytics in various sectors. Data centers are highly energy-intensive, with projected rack-level loads approaching ~1 MW by 2028; this is equivalent to the electrical power draw of 1000 US households going into a computing unit the size of a bookshelf. All of this energy ends up as waste heat, which needs to be removed to avoid overheating of electronics, thereby ensuring reliability and performance. Accordingly, significant water can be required onsite for cooling (direct use); this is in addition to the water requirements associated with electricity generation (indirect use) (Neudorfer and Llera, 2025; Bluefield Research, 2025; Amazon.Inc, 2020; Texas Water Development Board, 2021; D. C. Applications, 2025). Importantly, many upcoming data centers in the US are expected to generate their own power; many of them using natural gas. Water consumption associated with power generation for such cases cannot be classified as indirect use, but would count as direct use for regulatory and reporting requirements.

The data center energy–water tradeoff, governed by the chosen thermal management strategy, is characterized by Power Utilization Effectiveness (*PUE*) and Water Utilization Effectiveness (*WUE*, L/kWh),

defined as ratios of total facility energy and freshwater use to IT energy consumption, respectively (Azevedo et al., 2011; Green Grid, 2012; Fukuda et al., 2025). Direct evaporative cooling achieves low *PUE* (~1.3) but high direct *WUE* (1.7–2.5 L/kWh), corresponding to 2–19 million L per day of onsite water demand for mid to hyperscale data centers (Yañez-Barnuevo, 2025). Air-cooled systems have lower *WUE* (~0.2 L/kWh) at the expense of higher *PUE* (1.2–1.8), thereby increasing indirect freshwater demand (Shehabi et al., 2016; Shehabi et al., 2024). Data center water consumption is projected to increase 2–4-fold by 2028, reaching 150–270 billion L yr^{-1} in the U.S. alone (Tozzi and Analyst, 2025; Lei et al., 2025). Off-grid facilities with onsite gas-turbine generation will exacerbate scarcity due to elevated *WUE* (0.8–4.2 L/kWh) (COMPASS Research Consortium, 2026). Such water demands, often occurring in water stressed regions, present significant technological and environmental challenges for sustainable data center development. The present group has conducted an extensive analysis on various cooling techniques, power demand and associated water footprint of next-generation data centers, which has been documented in a white paper (COMPASS Research Consortium, 2026).

The quality of freshwater needed for both cooling and power generation is as important a consideration as the quantity. Freshwater needs to have < 500 ppm total dissolved solids (TDS), among other criteria (Tomberlin et al., 2020; Condair, 2020; Tomberlin et al., 2018). Water-associated hurdles impeding data center buildout would be greatly alleviated if saline water could be desalinated to meet the standards required for data centers. Fig. 1 shows a system-scale description of saline water desalination along with associated inlet-outlet saline streams and energy sources. Potential desalination feedwaters include brackish water (BW), seawater (SW), and industrial brines. BW (1000–10,000 ppm TDS) is typically sourced from inland aquifers and estuaries (Bruun et al., 2016), while SW (~35,000 ppm TDS) is the dominant global saline water resource (Elimelech and Phillip, 2011; Banchik et al., 2014). Industrial wastewater streams such as shale oilfield produced water (PW)—exhibit salinities of 50,000 to >200,000 ppm TDS (Survey et al., 2024). Although hypersaline PW presents significant environmental and disposal challenges, it offers valorization potential for freshwater recovery and also circular resource economy due to its mineral-rich composition (Katakam and Bahadur, 2024; U.S. Government Accountability Office (GAO), 2023; U.S. Department of Energy (DOE), 2024). Additional wastewater sources include textile effluents, mining brines, and power-plant blowdowns.

Desalination technologies (Fig. 1) are broadly classified as membrane-based or thermally driven processes. Membrane-based processes such as Reverse osmosis (RO) are pressure-driven (from electrical input to mechanical pumps), energy-efficient and cost-effective. However, they cannot effectively treat hypersaline brines (>50,000 ppm TDS) with substantial freshwater recovery, due to burst-pressure limit of the membrane, fouling, and membrane degradation (Bartholomew et al., 2017; Chen and Yip, 2018). Thermally driven processes such as Mechanical Vapor Compression (MVC) and Multi-stage flash (MSF) are typically feed salinity-insensitive and suitable for hypersaline desalination applications. However, such methods are very energy-intensive (compared to membrane desalination) due to the phase change based separation approach (Atia et al., 2021). Solar-driven interfacial evaporation (SIE) is a recently developed energy-efficient, renewable-powered desalination technique that uses a photothermal absorber at the air–water interface to drive localized evaporation of high salinity brine, yielding freshwater upon vapor condensation. Recent biomimetic and structurally engineered evaporator designs such as mangrove-inspired evaporator (Ding et al., 2025), a modular gradient-porous evaporator (Ding et al., 2026), a tree-inspired biomimetic evaporator (Ding et al., 2026), or a 3D wettability-patterned architecture (Ding et al., 2026), have addressed SIE's long-standing salt accumulation limitation. Overall, these advances substantially improve SIE's operational stability till zero-liquid discharge limits and salt-management practicality.

Fig. 1 also shows different desalination brine management strategies, which are a critical determinant of the environmental impact of desalination systems. Conventional discharge (~70,000 ppm TDS) (Moon et al., 2025) to receiving water bodies poses ecological risks. Minimum Liquid Discharge (MLD) concentrates brines to ~130,000–200,

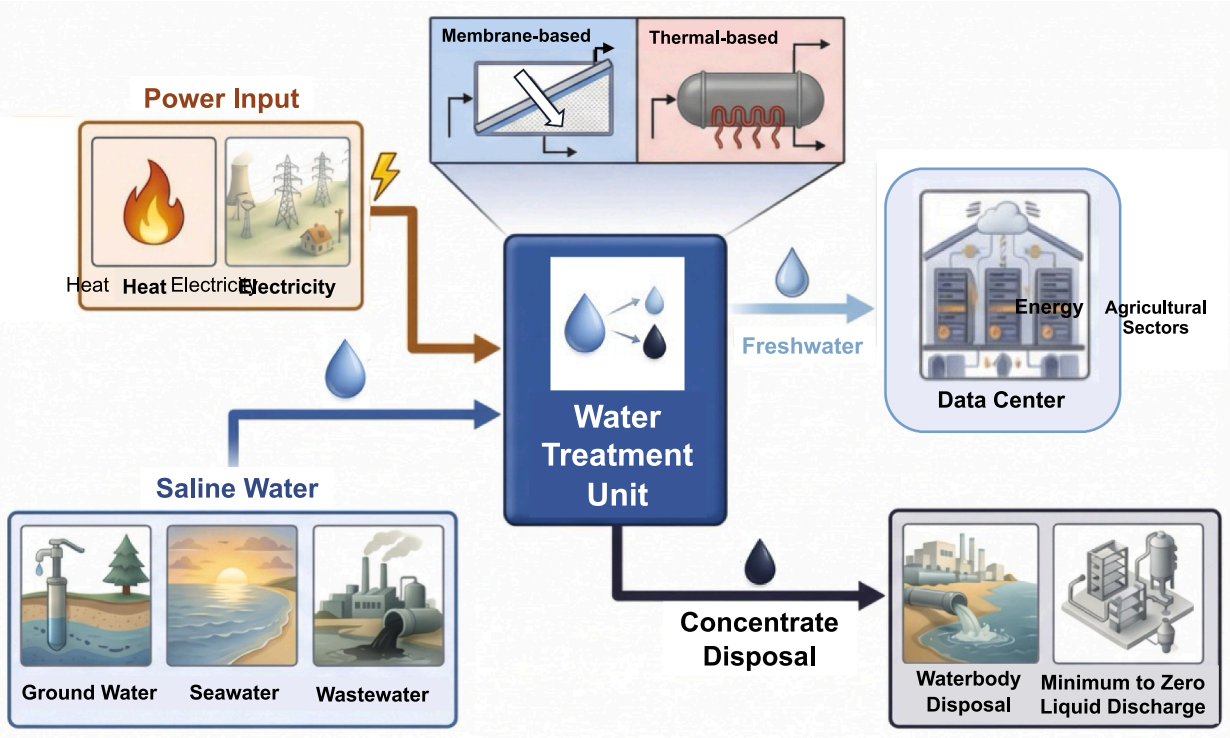


Fig. 1. Schematic illustration of desalination systems producing freshwater for data centers, along with depictions of feed streams, energy inputs, treatment pathways and brine disposal strategies.

000 ppm, thereby minimizing disposal volumes and reducing contamination potential (Zhang et al., 2025; Mo et al., 2022; Peters and Hankins, 2020). Zero-Liquid Discharge (ZLD) further concentrates brines toward saturation (~260,000 ppm NaCl), virtually eliminating liquid effluents while facilitating downstream mineral recovery (Thiel et al., 2015; Beni et al., 2022), an approach particularly attractive for inland, water-stressed regions (DuChanois et al., 2023; Nikkhah et al., 2024).

Multiple recent reports from US National Labs and industrial agencies have focused on the rapidly increasing energy and water demand metrics of data centers over the next decade (Shehabi et al., 2024; Aljbour et al., 2024), providing *PUE* and *WUE* projections and emphasizing novel cooling architectures, low-grade heat recovery and environmental impacts (Fukuda et al., 2025; Van Geet and Sickinger, 2010; Cejudo et al., 2023). **However, there is very limited analysis/discussion on the potential impact of saline water sources (via desalination) in meeting demands of data centers**. The few existing studies on desalination for data center applications (Yuan et al., 2023), restrict focus on waste-heat integrated, thermally driven processes (Kanbur et al., 2020; Sondur et al., 2018; Wyon, 2025; Chen et al., 2026) or thermoelectrically coupled RO systems (Yang et al., 2026) vulnerable to membrane performance degradation under high thermal loads (DuPont Water Solutions, 2022; DuPont Water Solutions, 2025; Karami et al., 2024). The low-grade heat and transient supply-demand mismatch inherent to waste heat integration, also require additional heat exchangers, heat pumps, or efficient thermal energy storage that could increase overall system cost, complexity and reliability risk compared to conventional operation (Yuan et al., 2023; Ebrahimi et al., 2014; Jagdale and Aljbour, 2024). These studies further rely on simplified first-principle models with process, property (Kanbur et al., 2020) or salt rejection assumptions (Yang et al., 2026) and area-normalized constant flux calculations (Kanbur et al., 2020; Yang et al., 2026), limited feed salinity ranges (Kanbur et al., 2020; Wyon, 2025; Chen et al., 2026) with negligibly low (Kanbur et al., 2020; Wyon, 2025) or singular recovery conditions (Chen et al., 2026; Yang et al., 2026), with such limitations in desalination modeling also identified as a substantial research gap in multiple recent studies (Atia et al., 2023; Bartholomew et al., 2018; Bartholomew et al., 2020; Dudchenko et al., 2021; Tucker et al., 2024). Furthermore, technoeconomic analysis of desalination for data center applications are not comprehensive and often exclude pretreatment, brine disposal or posttreatment costs tied to data center-specific water quality standards (Kanbur et al., 2020; Wyon, 2025; Chen et al., 2026; Yang et al., 2026).

Critically, bidirectional impacts of desalination performance and data center expansion— through which saline water utilization in desalination processes can influence the regional water scarcity beyond freshwater footprint alone – also remains largely unexplored, despite its relevance to sustainable siting decisions. Specifically, the influence of desalination-specific parameters such as freshwater recovery, feed salinity and brine disposal strategy, on the overall water stress impact of data centers needs to be quantified, as does the converse influence of data center specific thermal management operation (via *WUE*) on freshwater footprint, and upstream water withdrawal. Overall, significant research gaps remain in the techno-economic and environmental assessment of industrially mature desalination systems tailored for data center applications across representative feed salinities and concentrate handling strategies.

The initial benchmarking benefit is also pertinent from a fundamental system-level desalination modeling perspective. Previous MVC studies (Onishi et al., 2017; Onishi et al., 2017; Randon et al., 2020) were generally limited to a single feed concentration and recovery condition, despite MVC's suitability across a broad operating range;. Tucker et al. (Tucker et al., 2024) addressed this by reporting cost-optimal MVC performance across hypersaline produced-water concentration range up to ZLD limits, though, their framework used an overestimated evaporator heat transfer mean temperature difference and did not capture the variation in capital costs with operating conditions and boiling-point elevation (El-Dessouky and Ettouney, 2002). Studies of system-scale modeling of RO-Thermal hybrid desalination systems (Panagopoulos, 2021; Panagopoulos, 2022; Panagopoulos, 2022; Alsalal et al., 2026; Caceres Gonzalez and Hatzell, 2024), similar in scope to present study likewise involve simplified assumptions – arithmetic averaged stream-wise osmotic pressure and constant evaporator heat transfer coefficients (Caceres Gonzalez and Hatzell, 2024),single feed and recovery condition without cost-optimization (Alsalal et al., 2026), and or assumptions that are largely independent of feed concentration and recovery (Panagopoulos, 2021; Panagopoulos, 2022; Panagopoulos, 2022). Nearly all of these studies also use techno-economic cost parameters without inflation adjustment, underpredicting present-day water-treatment costs by neglecting capital-cost escalation and time value of money.

The present study addresses the above-mentioned research gaps via a detailed system-level design and technoeconomic evaluation of desalination techniques for data center freshwater production, holistically integrating pretreatment, post-treatment and concentrate management costs – each updated for present-day costs. Conventional RO, MVC, and its hybrids are modeled and techno-economically optimized for representative brackish, seawater, and hypersaline produced water treatment across conventional, MLD and ZLD concentrate disposal conditions, with freshwater recovery, specific energy consumption, second-law efficiency, and water cost (including pretreatment and posttreatment costs) quantified for each scenario. From a data-center perspective, this is the first study to provide such a detailed process, thermodynamic and economic analysis of these desalination technologies across the entire brackish-to-hypersaline feed range and conventional to MLD/ZLD brine management conditions. Importantly, the present study extends this analysis via a novel performance parameter, Saline Water Utilization Intensity, *SWUI* (L/kWh), which links desalination recovery and data-center water demand to regional water-scarcity implications. *SWUI* is evaluated for three representative data center deployment case studies till 2030, capturing the combined effect of cooling technique, local saline water conditions, geographic water stress, and brine disposal strategy on water availability.

Collectively, this establishes a desalination-centered framework — spanning industrially mature RO, MVC, and hybrid technologies, recovery-dependent thermodynamic and economic performance, and inflation-adjusted cost accounting relevant data centers— that goes beyond prior studies restricted to data-center water-use projections or waste-heat-driven, technologically nascent desalination configurations. By linking desalination performance to regional water-scarcity implications, the resulting *SWUI*-based framework further enables identification of application-specific trade-offs to guide saline-source selection and brine-management strategy, providing a standard benchmark for comparing future desalination technology developments, and lower-water-demand data-center designs.

It should be noted that the present study mainly involves comprehensive design and techno-economic analysis of RO and MVC techniques, with system-level integration referring to the coupling of the complete desalination treatment train, designed and cost-optimsed to data-center-specific freshwater quality (<500 ppm TDS) – with an evaluation of the resulting translational impact on regional water stress through the *SWUI* framework. Further system-level thermal or operational integration between the desalination system and the data center is beyond scope of the present study. As more detailed, facility-specific operational data for the computing infrastructure – capturing actual water and energy demand – become available for hyperscale deployments, they will extend the present system-level integration from both a mass- and energy-coupling perspective and improve the translational water-stress impact evaluation models for data centers.

2. Modeling and methodology

While many emerging desalination technologies can potentially be

used to meet the water needs of data centers (Ding et al., 2026; Atia et al., 2023; Peters and Hankins, 2019; Davenport et al., 2020), this study examines commercially available RO and MVC processes, noting their technical maturity, scalability, and availability of data for techno-economic analysis. Significantly, both these techniques provide a wide range of desalination capabilities with energy-efficient operation compared to other techniques within the respective membrane and thermal based categories (Thiel et al., 2015; Tucker et al., 2024). Fig. 2 shows a simplified schematic of two-stage RO and MVC with two-stage compression, whose working principles and modeling frameworks are detailed in Sections Sections 2.1.1 and 2.1.2, respectively. Prior to establishing individual modeling frameworks, fundamental stream and solute mass conservation equations are derived, consistent with previous studies (Katakam and Bahadur, 2025; Sree, Katakam and Bahadur, 2026; Katakam et al., n.d.).

$$\dot{m}_{fi} = \dot{m}_p + \dot{m}_{fe} \tag{1}$$

$$\dot{m}_{fi}(c_{fi}) = \dot{m}_p(c_p) + \dot{m}_{fe}(c_{fe}) \tag{2}$$

where, $\dot{m}_{fi,p,fe}$ and $c_{fi,\ p,\ fe}$ are the feed, permeate and feed exit stream mass flow rate (kg/s) and concentrations (ppm) respectively. It should be noted that Eqs. (1) and (2) are valid for mathematically formulating module, stage and overall system-level mass conservation, with the feed exit stream corresponding to desalination concentrate and post-treated permeate corresponding to freshwater stream for system-level analysis.

The freshwater recovery, *RR* (L-permeate/1000 kg-feed) for the overall desalination system can be expressed as:

$$RR = \frac{\dot{m}_p}{\dot{m}_{fi}} \times \frac{1000}{\rho_p} \tag{3}$$

where, ρ_p is the density (kg/m^3) of freshwater stream based on its solute concentration.

2.1. Design and thermodynamic analysis of RO and MVC desalination systems

2.1.1. Description of RO modeling framework

Reverse osmosis involves pressurized feed from hydraulic pumps and pressure exchangers (e.g., coupled turbine-pump) passing through a membrane stage with semi-permeable salt-rejecting elements. Hydraulic pressure difference, ΔP (Pa) exceeding transmembrane osmotic pressure generates water flux with minimal salt passage, producing a permeate stream. In a series-coupled two-stage RO system (Fig. 2a), dewatered brine from the first stage is further pressurized via a similar pump-pressure exchanger setup and redirected as feed to the second stage for additional permeate recovery. Permeate from both stages is post-treated to yield freshwater, while dewatered concentrate from the second stage is depressurized via pressure exchangers (PXs) and either disposed into waterbodies or further desalinated through MVC (RO-MVC hybrid) to achieve MLD/ZLD standards. Two-stage RO is considered over single-stage RO for its lower energy consumption and treatment costs, achieved through optimized inter-stage permeate recovery distribution enabled by flexible control over stage-level transmembrane pressure difference (Thiel et al., 2015; Anvari et al., 2024; Wei et al., 2017). The average permeate concentration, c_{pavg} for each stage with a given recovery *RR* and c_{fi} can be obtained from:

$$c_{p_{avg}} = \frac{c_{fi}}{RR}\left(1-\left(1 - RR \times \frac{\rho_{pavg}}{1000}\right)^{(1-MRR)}\right) \tag{4}$$

In Eq. (4), membrane rejection ratio *MRR* is expressed as:

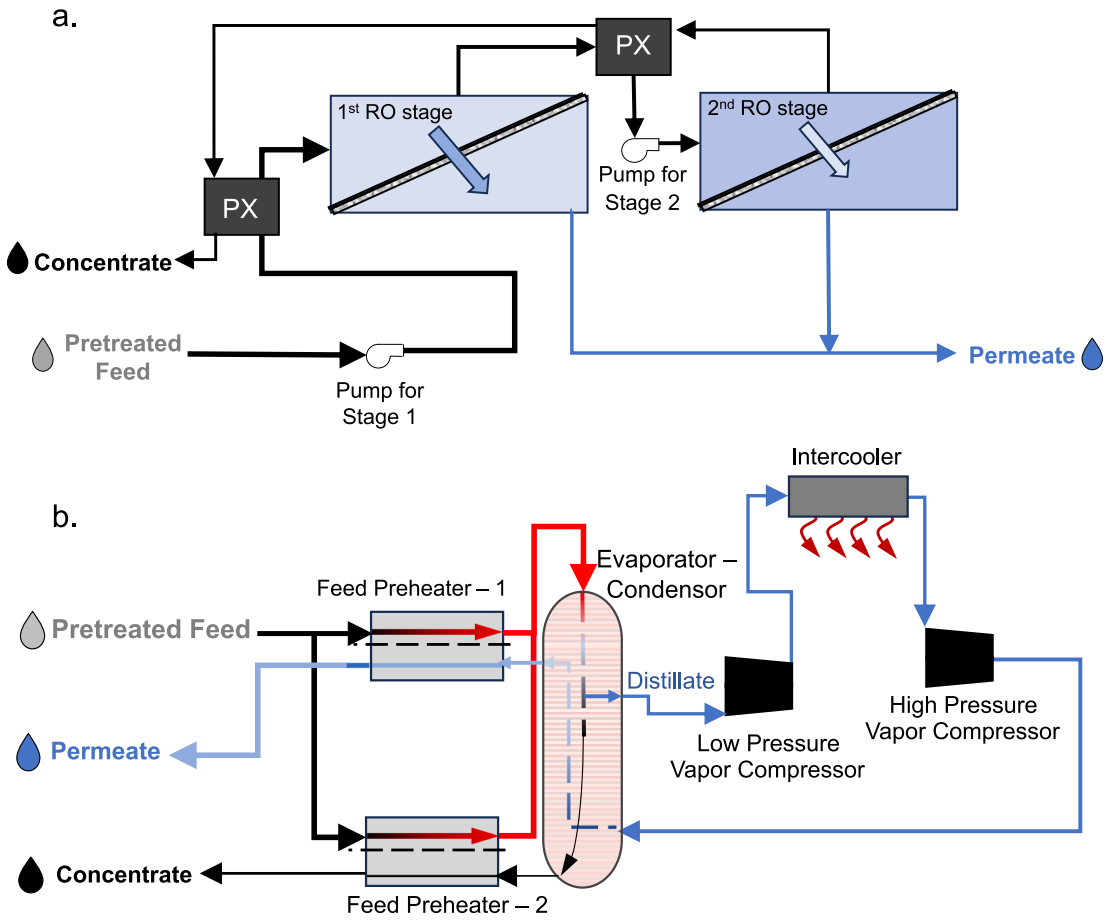


Fig. 2. Schematic illustration of a) two-stage RO coupled in series b) MVC with two stage compression.

$$MRR = 1 - \frac{\lim_{RR\to 0} c_{p_{avg}}}{c_{fi}} \tag{5}$$

The required membrane area A_m (m^2) for each stage based on corresponding feed inlet conditions, recovery and applied ΔP is determined using a recently developed heat-mass transfer analogy-based method (Katakam and Bahadur, 2025):

$$A_m = \frac{C_{fsolv}}{A\rho_{psolv}} \ln\left(\frac{J_{w1}}{J_{w2}}\right) \frac{1}{(1-C_r)} \tag{6}$$

The detailed derivation and implementation methodology of Eqs. (4)-(6) can be found in our previous study (Katakam and Bahadur, 2025), which also includes substantial corroboration for the membrane rejection ratio based modeling approach for RO stage design. In Eq. (6), C_{fsolv} refers to solvent capacity of feed stream (kg/(s-Pa)), $J_{w1,2}$ refers to the transmembrane pure water flux at the feed inlet and exit of the RO stage respectively, ρ_{psolv} is the pure water density (kg/m^3) at process conditions, C_r is the ratio of feed to permeate solvent capacity, A is the water permeability coefficient (m^3/m^2-s-Pa). The combined specific work input, W_{inp} (kWh/m^3-freshwater) for two-stage RO system is determined by proportional contribution of depressurization flow work of second stage feed exit stream based on power requirement for each RO stage without pressure exchangers:

$$W_{inp} = \left(\Delta P_1 - (\alpha)\Delta P_2 \eta_{PX}\left(\frac{(1-RP_2\ )(1-RP_1)\rho_{fi1}}{\rho_{fe2}}\right)\right) \frac{\rho_p}{(3.6\ \times 10^6)\rho_{fi1}(RP_1 + RP_2(1-RP_1))\eta_P} + \left(\Delta P_2 - \Delta P_1 - \Delta P_2(1-\alpha)\eta_{PX}\left(\frac{(1-RP_2\ )\rho_{fe1}}{\rho_{fe2}}\right)\right) \frac{\rho_p\ (1-RP_1)}{(3.6\ \times 10^6)\rho_{fe1}(RP_1 + RP_2(1-RP_1))\eta_P} \tag{7}$$

$$\alpha = \frac{\frac{\Delta P_1}{\rho_{fi1}}}{\frac{\Delta P_1}{\rho_{fi1}} + \frac{(\Delta P_2 - \Delta P_1)(1-RP_1)}{\rho_{fe1}}} \tag{8}$$

For Eqs. (7) and (8), subscripts 1 and 2 refer to specific parameters for first and second RO stages respectively, $RP = \frac{\dot{m}_p}{\dot{m}_{fi}}$ refers to permeate recovery ratio, $\eta_{PX,P}$ are pressure exchanger and pump efficiencies respectively, $\rho_{fi,fe}$ are the density of feed inlet and exit streams for referred stage. To conservatively incorporate the effects of fouling and viscous effects along with circulating pump energy consumption, an additional 5% of work input and membrane area are considered. These miscellaneous parasitic energy effects can be further mitigated using energy recovery devices (ERDs) aimed at reducing overall power consumption (Swidan et al., 2024; Energy Recovery, Inc). In addition, a mean and maximum deviation 2.5% and 4.8% when observed for comparing W_{inp} results from the present method against system-specific circulating pump power inputs calculated with viscous losses. This was obtained by considering a typical averaged feed side pressure drop of 0.5–3 bar (Bartholomew et al., 2018; Bartholomew and Mauter, 2019; Mistry et al., 2011), across brackish, seawater and 50,000 ppm feed conditions, with RP ranging from 30% to maximum limit (corresponding with concentrate stream concentration close to 75,000 ppm). For a similar stage-level pressure drop of 0.5–3 bar, <0.06 kWh/m^3-permeate of minimal net deviation in W_{inp} was observed for the cost-optimal BW and SW RO 2 stage systems reported in Section 3.1.

2.1.2. Description of MVC modeling framework

Fig. 2b shows a system-level description of thermally driven MVC, wherein saline feed is preheated to its phase-change temperature and dewatered through vaporization in the evaporator-condenser unit, generating concentrated brine and superheated distillate vapors with negligible salt concentration $(c_p = 0)$ (Thiel et al., 2015; Tucker et al., 2024; El-Dessouky and Ettouney, 2002). The distillate stream is compressed through two-stage compressors and redirected as condensate, delivering condensation latent heat for feed vaporization. Condensed distillate and concentrated brine at the evaporator exit are sensibly cooled through feed preheaters, yielding freshwater and concentrate streams. The specific work input, W_{inp} (kWh/m^3-permeate) and heat exchanger areas, A_{HX} (m^2) for MVC two-stage compression, along with relevant thermophysical properties are determined based on existing models (Thiel et al., 2015; El-Dessouky and Ettouney, 2002; Swaminathan et al., 2016):

$$W_{inp} = C_{pv}\frac{273 + T_{ci}}{\eta_{comp}}\left(PR_1^{\frac{\gamma-1}{\gamma}} - 1\right) \times \frac{\rho_p}{3.6\times 10^6} + C_{pv}\frac{273 + T'_{ci}}{\eta_{comp}}\left(PR_2^{\frac{\gamma-1}{\gamma}} - 1\right) \times \frac{\rho_p}{3.6\times 10^6} \tag{9}$$

$$A_{HX} = \frac{\dot{Q}}{U_{HX}(LMTD_{HX})} \tag{10}$$

For Eqs. (9) and (10), η_{comp} is the compressor isentropic efficiency, C_{pv} is the specific heat capacity of water vapor (kJ/kg-K) at constant pressure, T_{ci} and T'_{ci} are the compressor inlet fluid temperatures of superheated distillate at first and second stage respectively, γ is the heat capacity ratio, $PR_{1,2}$ are the pressure ratios of first and second stage compression respectively, $\dot{Q}$ is the rate of heat transfer (W) in the heat exchanger, U_{HX} is overall heat transfer coefficient (W/m^2-K), and $LMTD_{HX}$ (K) is the logarithmic mean temperature difference for heat transfer at inlet and exit of heat exchanger (Cengel and Ghajar, 2011). The pressure ratios for each stage is determined based on well-established compression with interstage cooling optimization approach (Ganesan, 2010). An additional 5% of W_{inp} and A_{HX} for every heat exchanger is assumed beyond that calculated by established MVC design methods, to conservatively incorporate viscous and heat exchanger fouling effects. Accordingly, the mathematical expressions for $PR_{1,2}$ and the parameters describing heat transfer in evaporator-condenser and feed preheaters from Eq. (10) are provided in Section S.1. of Supplementary Information. Compressor pressure ratios are determined based on evaporator-condenser terminal temperature difference (TTD), an input parameter defining the required temperature difference between condensate and evaporator streams for feasible heat transfer. Superheated distillate is compressed to higher pressure such that the corresponding saturation temperature for condensation exceeds the evaporation phase-change temperature of the feed, enabling thermodynamically feasible heat transfer. Higher TTD results in higher overall compression pressure ratios for superheated distillate—increasing W_{inp} but decreasing A_{HX} (Eq. (10)) as the driving force for evaporator heat transfer increases. The opposite trend occurs as TTD approaches 0 K, highlighting TTD as a critical design parameter that simultaneously influences W_{inp} and A_{HX}, suggesting an optimal

operating point exists for minimizing overall water treatment cost.

2.2. Technoeconomic modeling

The present study defines an overall specific water cost (WC) ($/m^3-permeate) for a desalination technique, by including the annual capital (ACE) and operating expenditure (AOP) normalized by the annual freshwater production as:

$$WC = \frac{CF(ACE) + AOE}{AWP} \tag{11}$$

In Eq. (11), CF refers to the amortized capitalization factor, dependent on plant life and interest rate. For membrane-based RO, WC is calculated using the framework developed by Mo et al. (Mo et al., 2022), which includes capital expenditure for membranes, pumps, pressure exchangers, and feed pretreatment, while operational expenditure comprises electricity, membrane replacement, chemical, maintenance, labor, and concentrate disposal/post-treatment costs. Furthermore, capital and operating expenditure for permeate posttreatment is also included based on results from Santos et al. (dos Santos et al., 2024). An additional 10% of membrane replacement costs accounts for accelerated membrane deterioration due to fouling. Techno-economic optimization of the two-stage RO process via WC minimization is iteratively determined across all feasible stage-level freshwater recovery and transmembrane pressure combinations, incorporating non-ideal solution behavior, membrane selectivity, and concentration polarization effects. This approach contrasts with previous studies (Thiel et al., 2015; Anvari et al., 2024; Wei et al., 2017) that minimize only W_{inp} or A_m based on simplified property and process assumptions, or constrained design parameters to obtain optimal two-stage RO configurations.

Water treatment cost for MVC is determined from models provided by Swaminathan et al. and Bartholomew et al. (Bartholomew et al., 2018; Swaminathan et al., 2016), where ACE comprises feed pretreatment (excluded for MVC treatment cost estimation in RO-MVC hybrid analysis), preheater, compressor, and evaporator costs (including pressure drop), while AOE comprises electricity, maintenance and labor, chemicals, and concentrate disposal/post-treatment. Optimal MVC configuration for given desalination conditions and evaporator-feed inlet concentration is determined for a range of terminal temperature difference TTD values for the evaporator-condenser. Additional system integration cost of 10% of ACE is included for both RO and MVC systems, while an additional 10% of evaporator and preheater capital expenditure is included to account for heat exchanger fouling effects.

2.3. Modeling impact of data center water requirements

While conventional performance metrics such as WUE, water cost, and specific energy consumption assess operational effectiveness of thermal management and desalination systems from a data center water demand perspective, no metric directly quantifies the practical implications of data center water footprint on regional water scarcity. Additionally, apart from the parameters mentioned, feed characteristics and water treatment performance of the desalination systems associated with data centers influence the site-specific saline water withdrawal requirements, thereby affecting local water availability. These factors necessitate a comprehensive analysis on the water-related impacts of data centers, for enabling realistic environmental assessments and sustainable siting of computing infrastructure. Although the Water Utilization Intensity (WUI) (Neudorfer and Llera, 2025; Jiang et al., 2025) incorporates data center water stress implications, it is defined solely based on freshwater WUE, excludes site-dependent freshwater supply and desalination operational factors, and has been calculated exclusively based on evaporative cooling strategies, which are no longer being considered in any significant manner. In addition, this parameter is not evaluated for varying water stress projections under future climate change or development scenarios. Therefore, the influence of desalinated saline water sourcing, brine management and evolving regional water stress factors on future data center water sustainability remains largely unexplored. To provide an improved, comprehensive assessment capturing multiple factors governing data center and desalination operations, this study introduces Saline Water Utilization Intensity, $SWUI$ (L-feed/kWh) as:

$$SWUI = \frac{WUE \times 10^6}{RR \times \rho_{fi}}(WS + 1) \tag{12}$$

$SWUI$, as formulated in Eq. (12), is the amount of saline water withdrawn for desalination per unit IT energy consumption, multiplied by a parameter directly proportional to local water stress value. At a more detailed level, the first term on the right-hand side (outside brackets) of the equation represents the saline water utilization effectiveness, $SWUE = \frac{WUE \times 10^6}{RR \times \rho_{fi}}$ (L-feed/kWh), of the data center infrastructure, which is determined by both the deployed thermal management technique (determines WUE) and the desalination technique (captured by RR, which depends primarily on feed inlet condition C_{fi}, ρ_{fi}, and the concentrate disposal/post-treatment strategy). This term essentially provides the ratio of saline water demand per unit IT energy consumption. Saline water becomes the actual commodity that is directly extracted from the nearby environment instead of freshwater. Accordingly, $SWUE$ and $SWUI$ terms are quantified in Eq. (12), to capture the effects of saline water demand on ground, surface, or unconventional water sources. In order to capture facility-level water footprint of the computing infrastructure and its impact on local water stress, mainly direct WUE values are considered for calculating $SWUI$ from Eq. (12).

The second term (inside brackets in Eq. (12)) includes regional water stress (WS) determined from the Aqueduct Water Risk Atlas (Temporal resolution Unimproved /No Drinking Water Analyze Click map Enter Address, 2023), defined as the ratio of projected water demand across industrial, domestic, agricultural, and other sectors to projected renewable water resource availability based on local weather patterns and global warming effects. WS ranges from 0 to 5, where 0 indicates negligible stress and 5 signifies severe stress. A value of 1 is added to WS in the equation, consistent with WUI formulation (Neudorfer and Llera, 2025), to avoid improbable zero $SWUI$ results. It should be noted that, WS term in Eq. (12) is not included to represent stress on the saline feed source itself. Rather, WS represents the regional water stress context in which the desalination-supported data center operates. The freshwater ultimately produced through desalination, regardless of feed source, is allocated to data center operation within a region where it could alternatively satisfy competing municipal, agricultural or industrial water demands. Accordingly, $SWUI$ in Eq. (12) is formulated as a first-order water utilization intensity metric that quantifies the relative water-utilization implications of deploying a given desalination-supported data center under different regional water stress conditions, rather than the depletion or intrinsic sustainability of the saline feed source itself.

$SWUI$ in Eq. (12) is formulated to increase monotonically with both desalination-related saline water withdrawal (via $SWUE$) and regional water stress (WS), reflecting the combined impact of data-center deployment and its associated desalination operations in more water-stressed regions. A linear proportionality of $SWUI$ to both the parameters is considered because firstly, $SWUI$ is not a directly measurable physical quantity for which experimental data or regression analysis could justify a specific nonlinear relationship; imposing a polynomial or other nonlinear form would add arbitrariness rather than remove it. Second, the linear form was intentionally kept consistent with previously reported WUI-based metrics for data centers, enabling direct comparison with the existing WUI metric (as discussed in Section 3.3) while extending it to explicitly account for desalination-based water supply.

Overall, $SWUI$ is a practical, easily applicable and unified metric which captures the integrated effects of upstream desalination, available

saline water resources and data center thermal management techniques on local water stress. Section 2.4 details case studies involving three representative hyperscale data centers in the US, for which *SWUI* values are calculated using *WUE* and commercial desalination technologies. Section 3.3 provides further discussion on the benefits of adopting the proposed *SWUI* for evaluating water related implications for data centers over existing analysis frameworks. In order to evaluate the repeatability of *SWUI* trends for linear, quadratic and exponential dependency across all the case studies analyzed, a sensitivity analysis for each formulation is provided in Table S.2 of Supplementary Information and discussed further in Section 3.3. A detailed derivation of Eq. (12) is provided in Section S3. of Supplementary Information.

2.4. Methodology and input parameters

Since >93–97% of dissolved salt in pretreated brackish, sea, and produced water is aqueous NaCl, the water chemistry, concentration, and overall design of the analyzed desalination systems are considered based on NaCl solutions (Atia et al., 2021; Thiel et al., 2015; Beni et al., 2022) as inlet and outlet streams. This is a reasonable approach considered and supported by multiple recent studies, wherein pre-treatment of the saline feed results in almost negligible impact of fouling, scaling and other associated effects on desalination performance (Atia et al., 2023; Tucker et al., 2024; Turetta et al., 2026; Al-Maas et al., 2025). Typical pretreatment includes ultrafiltration, nanofiltration, and coagulation-flocculation, while common permeate post-treatment methods—UV sterilization and electro-deionization, also used in semiconductor applications (Mulder, 1996)—are considered for enhancing freshwater quality.

It should be noted that while the present study provides a system-scale assessment of RO, MVC, and hybrid desalination performance, certain aspects warrant further investigation. Future work should incorporate more representative industrial feed compositions containing multivalent ions, sulfates, silica, and other scaling precursors to better evaluate real-world desalination performance. The objective of the present study is to provide a system-scale assessment of desalination performance, noting that the selection and optimization of specific pretreatment strategies are highly dependent on feedwater composition and site-specific operating conditions. Future studies could systematically correlate fouling behavior, scaling propensity, flux decline etc. while evaluating the associated impacts on specific pretreatment requirements. Such analyses may necessitate the inclusion of additional pretreatment processes, including chemical softening, pH adjustment, antiscalant dosing etc. particularly for high-recovery RO and MVC systems.

For brackish and seawater feeds, two-stage RO is considered for desalination to conventional disposal limits (Table 1.), with further concentrate dewatering to MLD and ZLD standards via coupled MVC (hybrid RO-MVC), as described in Section 2.1. Given RO's limited suitability for treating hypersaline produced water (≥50,000 ppm) due to membrane pressure constraints and other factors, MVC desalination is considered for such concentrations under conventional disposal, MLD, and ZLD scenarios.

Based on inlet feed concentration and concentrate disposal/post-treatment strategies, Eqs. (1)–(5) are solved to determine flow rate, overall *RR*, and outlet stream concentrations (permeate and concentrate) for RO processes, as detailed in Section 2.1.1. For MVC processes, assuming negligible permeate salt concentration, Eqs. (1)–(3) directly yield outlet stream properties and freshwater recovery. For each desalination scenario, W_{inp} and membrane area (RO) or heat exchanger area (MVC), along with associated operating variables, are determined across a range of design parameters: stage-level freshwater recovery $\left(0 < RP_{1,2} < RP;\ RP = RP_1 + (1 - RP_1)RP_2\right)$ and pressure distribution $(0 - \Delta P_{max})$ for two-stage RO, and *TTD* (0–5 K) for MVC (Thiel et al., 2015; Wei et al., 2017). Individual stage pressure requirements for corresponding stage

Table 1
Key design, operation and cost parameters in this study.

Process parameters		
Parameter	Value	Reference
Feed inlet flow rate	200 m^3/hr	
Feed inlet concentration, c_{fi} (ppm)	5000 (brackish water) 35,000 (seawater) 50,000 (Mild Hypersaline PW) 91,000*-150,000 (Hypersaline PW)	(Banchik et al., 2014; Survey et al., 2024; Katakam et al., n.d.)
Feed stream inlet temperature	25 °C	
Permeate concentration limits	<500 ppm	(Tomberlin et al., 2020)
Concentrate stream concentration, c_c (ppm)	70,000 (conventional disposal#) 130,000–200,000 (MLD) 200,000–260,000 (ZLD)	(Moon et al., 2025)
Input parameters for RO&		
Water permeability coefficient, *A*	4.2×10^{-12} m^3/Pa-m^2-s	(Bartholomew et al., 2018; Lu et al., 2007)
Salt Permeability Coefficient, *B*	3.5×10^{-8} m^3/m^2-s	"
Membrane Rejection Ratio, *MRR*	0.99	(Katakam and Bahadur, 2025)
Maximum applied transmembrane pressure difference	65 bar	
Pump efficiency, η_P	80%	(Bouma and Lienhard, 2018)
Pressure Exchanger efficiency, η_{PX}	96%	"
Input parameters for MVC$		
Overall heat transfer coefficients for feed preheaters, $U_{ph1.2}$	1.18 kW/m^2-K	(Swaminathan et al., 2016)
Feed stream temperature at evaporator-condenser inlet, T_{eci}	50 K	(Thiel et al., 2015)
Terminal Temperature Difference (ΔT_{ttd})	0–10 K	(Tucker et al., 2024)
Compressor efficiency, η_{comp}	0.8	"
Input parameters for Technoeconomic Modeling @		
Load Factor, *LF*	0.9	(Beni et al., 2024)
Membrane Cost parameter, *CPmem*	48 $/$m^2$	"
Pump Cost parameter, *CPpump*	74 $/($m^3$-bar/hr)	"
Pressure Exchanger Cost equation, *ACCPX*	$4369.1\left(\frac{\dot{m}_{fe}}{\rho_{fe}}\right)^{0.58}$	"
Amortization interest rate, *r*	8%	"
Designed plant lifetime, *y*	20 years	"
System Integration Cost, *SIC*	10% (*ACC*)	(Kar and Bahadur, 2020)
Practical Investment Factor, *PIF*	1.4	(Mo et al., 2022)
Membrane replacement cost, *AOCmem*	11% (*CAPEX* of membrane)	
Electricity Cost, *CPelec*	0.14 $/kWhr	(Island, 2026)
Cost of preheated feed mixer, *CPmix*	503.2 ($/(L/s))	"
Intake and Pretreatment cost parameter, *CPInpre*	4418 $/($m^3$-feed/hr)	"
Concentrate handling cost parameter, *CPconc*	0.073$/($m^3$-concentrate) for conventional disposal; 0.44 $/($m^3$-concentrate) for MLD and ZLD post-treatment	"
Pressure Vessels cost	20% (*CAPEX* of membrane)	(Patel et al., 2024)
Posttreatment cost parameter, *CPpost*	0.41 $/$m^3$-permeate (capital) + 0.03 $/$m^3$-permeate (operational)	(dos Santos et al., 2024)

[*] Based on mean PW concentration obtained across 85 wells from USGS database for Permian Basin, Texas.

[#] Based on typical seawater RO brine salinity and maximum concentrate TDS limits obtained from conventional RO processes.

[&] Other relevant input parameters such as feed inlet conditions such as hydraulic diameter and feed inlet Reynolds number can be found at (Katakam and Bahadur, 2025).

[$] Thermophysical properties of aq. NaCl streams and distilled vapor along with emperical correlations for evaporator-condenser heat transfer coefficient are available at (Thiel et al., 2015; El-Dessouky and Ettouney, 2002).

[@] Other cost parameters and technoeconomic modeling framework not mentioned in the above table is adopted from Mo et al. (Mo et al., 2022) and Swaminathan et al. (Swaminathan et al., 2016), normalized to $\$_{2026}$ using CEPCI values, with detailed equations provided in Section S.4 of Supplementary Information.

recovery in two-stage RO are determined based on excess pressure ΔP_{ex} (0–30 bar) above maximum transmembrane osmotic pressure at the membrane-fluid interface across each stage. However, ΔP exceeding the 65-bar maximum applied transmembrane pressure limit for RO membranes (approaching membrane burst pressure) is considered infeasible and is excluded from subsequent techno-economic analysis. For each RO and MVC operating condition, corresponding capital and operational expenditures and finally *WC* are determined, enabling identification of techno-economically optimized conditions for minimized water costs, as detailed in Section 2.2. In addition to the design methods discussed in Section 2.3, 2nd law analysis of the cost-optimal desalination systems is also conducted in this study, with relevant information provided in Fig. S1 and Section S.2 of Supplementary Information. To avoid unreasonable desalination operating conditions, systems with a freshwater recovery of at least 100 L-permeate/1000 kg-feed are considered in the present analysis.

To quantify the translational impact of data center operations and desalination performance on regional water stress, *SWUI* (Section 2.3) is determined for three representative data center developments in the US in: i) Inland region with mild water stress ($WS < 3$) and with abundant brackish water availability, ii) Coastal region with substantial to severe water stress ($WS = 3-5$) and which will mainly depend on seawater desalination for obtaining freshwater, and iii) Semi-arid to arid regions with severe water stress levels ($WS > 4$) but with significant potential for produced water reuse and desalination, which preferably operates near ZLD standards to enable critical mineral extraction. Specific data center projects in Texas, which correspond to these cases include Project Rose in Longview (Case 1), Project Gladys near Corpus Christi (Case 2), Project Crusoe in Abilene (Case 3) (proximity to Permian Shale oilfield), respectively. The present study considers a feed with a TDS level of 91,000 ppm; this is the average TDS of produced water from the Permian Basin in Texas obtained from a USGS report[14]. This value was calculated based on measured TDS of brine from different wells, with concentrations ranging from 7000 ppm to >200,000 ppm. It should be noted that the present analysis can be used to evaluate the impacts of data center deployment at any location, if projected water stress levels, available saline water characteristics and brine disposal strategies are known. These specific case studies were considered for their distinct geographical, hydrological and saline water conditions, enabling *SWUI* evaluation for a wide range of representative desalination scenarios (relevant to data centers). This diversity provides *SWUI* results across the spectrum from mild to severe water stress impact, resulting in a wider assessment of water related aspects based on data center operations (related to *WUE*) and desalination metrics (such as freshwater recovery, feed salinity etc.) The most applicable desalination scenarios for each case study are therefore identified, alongside projected *WUE* and *WS* values, to obtain *SWUI* results. This analysis specifically focusses on scenarios involving planned or under-development hyperscale data center facilities.

Averaged *WS* values are determined under pessimistic, optimistic, and business-as-usual climate scenarios, while empirical correlations for

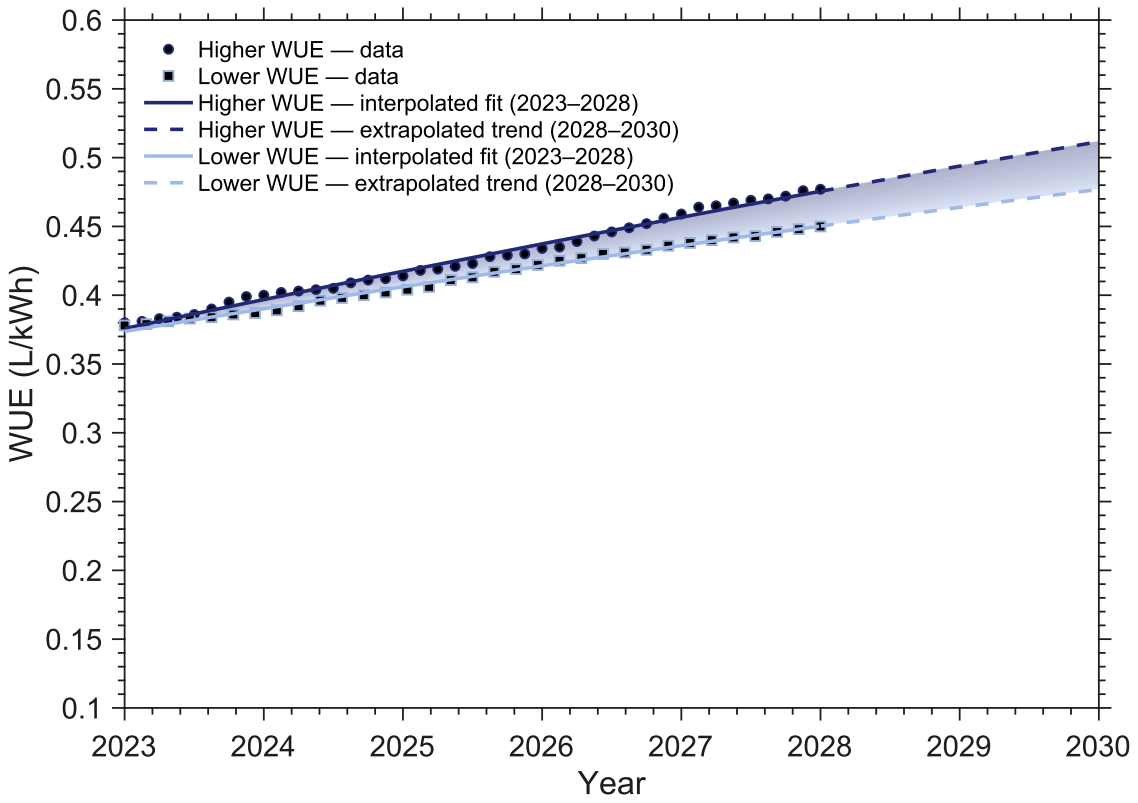


Fig. 3. Comparison of projected data, interpolated equations and extrapolated trends for higher and lower limit WUE values from 2023–2030, as per report by LBNL (Shehabi et al., 2024).

projected *WUE* averaged across different thermal management strategies through 2028 are obtained from the LBNL report (Shehabi et al., 2024), given by Eqs. (13) and (14). The equations are derived by interpolating reported data available through 2028 and subsequently applied to estimate *SWUI* up to 2030, involving minimal extrapolation of *WUE* to extend the analysis toward the end of this decade. Fig. 3 compares these *WUE* formulations with reported data (R^2>0.99 for both the interpolated equations), which were observed to increase in the near future due to increased shift towards hyperscale data centers.

It should be noted that the predicted *WUE* ranges obtained from the report are based on different values obtained for different data center deployment scales, seasonal weather variation, geographic location, operational practice uncertainty etc. among other aspects considered. Therefore, the lower and higher limits obtained based on the averaging across all these multiple data center operation and thermal management scenarios, is considered for its values are determined based on a holistic and rigorous simulation database. However, as mentioned previously, the models introduced in the present study can be utilized for analyzing any practically obtainable combination of data center, desalination and geographic parameters.

$$(\mathrm{WUE})_{\mathrm{high}} = 1.133738306 - 0.7579530345^{*}\exp\ (-0.02822012058\ (\mathrm{year} - 2023)) \quad (13)$$

$$(\mathrm{WUE})_{\mathrm{low}} = 0.8055832427 - 0.4319195671^{*}\exp\ (-0.03902866069\ (\mathrm{year} - 2023)) \quad (14)$$

Table 1 lists key input parameters and process conditions used in this study, other relevant input parameters can be obtained from references cited. The cost parameters are normalized to $\$_{2026}$ using Chemical Engineering Plant Cost Index (CEPCI). Equations related to techno-economic analysis for RO and MVC systems are provided in Section S.3 of Supplementary Information. CEPCI based cost-inflation for both the techniques and evaporator material-related effects specific to MVC were included in the analytical framework. Membrane property values similar to other recent system-level desalination modelling and pilot studies (Beni et al., 2022; Atia et al., 2023; Alsalal et al., 2026; Beni et al., 2024; Al-Amoudi et al., 2023) are considered in the present analysis

3. Results and discussions

3.1. Freshwater yield, energy and cost metrics

Based on input parameters in Table 1, freshwater recovery, W_{inp} and *WC* for different cost-optimal desalination systems designed across brackish to hypersaline feeds is shown in Fig. 4a–c., with the corresponding 2nd law efficiency results provided in Fig S2. of Supplementary Information. Brackish water feeds show higher freshwater yield of >900 L/kg-feed across different desalination scenarios due to low salt content in feed composition (Table 1). Recovery increases by about 6% from concentrate disposal to near-saturation ZLD disposal (260,000 ppm) due to higher brine concentration (Eqs. (1) to (5)). Recovery decreases with increasing feed concentration, especially from seawater towards hypersaline cases (75% reduction for MLD upper limit disposal) due to higher salt content in the feed, resulting in lower freshwater recovery needed for reaching specific brine disposal standards (Eqs. (1) to (3)). For the same reasons, MLD-based concentrate

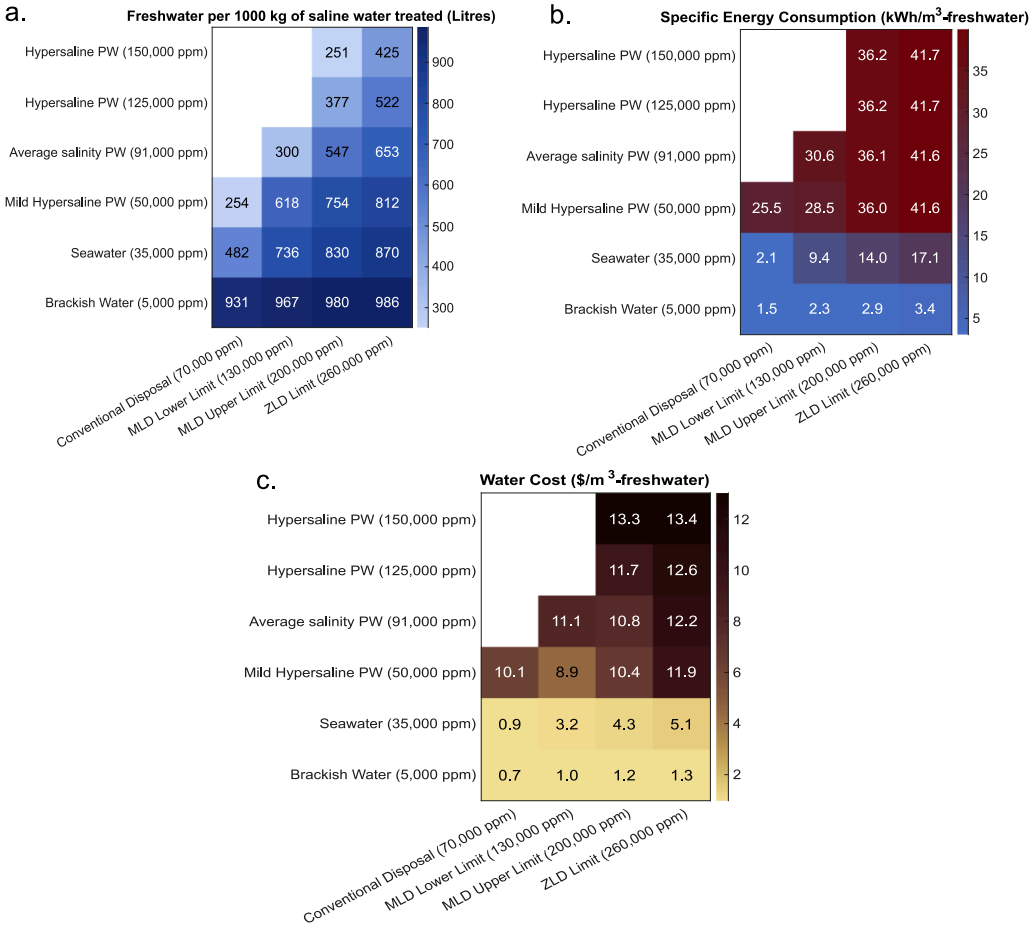


Fig. 4. Performance metrics of conventional RO and MVC desalination systems designed for different saline water treatment and brine disposal applications: a) Freshwater recovery, b) Specific Energy Consumption, and c) Water Cost.

management for hypersaline PW feed shows lowest recovery as minimal dewatering is needed to increase feed concentration from 125,000 - 150,000 ppm to the MLD standards of 130,000–200,000 ppm. Similarly, conventional brine disposal strategy for mild hypersaline PW feed also results in lower freshwater yields as shown in the figure due to minimal dewatering requirements needed for reaching 70,000 ppm concentrate from 50,000 ppm feed. The freshwater yield for these scenarios are >74% lower than the maximum *RR* observed for brackish water with ZLD upper limit disposal. It should be noted that such substantial decreases in freshwater recovery with respect to increasing feed inlet concentration and concentrate handling conditions, would result in increased saline water withdrawal by data centers. This would result in increased regional water stress from the data center facilities, depending on the local climate and hydrological conditions. Further discussion related to these aspects is provided in detail in Section 3.3, wherein *SWUI*-related results are discussed.

Fig. 4b shows that although W_{inp} metrics for brackish and seawater are substantially lower than that for hypersaline PW treatment, their energy consumption increases drastically (more than doubles for brackish water and increases eight-fold for seawater from concentrate disposal to ZLD limit) from concentrate disposal to ZLD scenarios. This is mainly due to the increased contribution of the energy-intensive MVC process needed to achieve MLD to ZLD standards, noting that RO cannot be used for such high salinities. Consequently, MLD and ZLD systems, while more sustainable, inherently demand higher energy than conventional brine disposal. For PW treatment of 50,000–150,000 ppm feed using thermal MVC across scenarios, the W_{inp} increase (1.2–1.6x from lowest reported W_{inp} to that of ZLD limit) is lower than for brackish and seawater due to the feed-insensitive operational advantage of thermal desalination (also observed in (Thiel et al., 2015)), which results in limited increase in energy demand even near ZLD limits. This explains the negligible W_{inp} increase between PW treatment cases for a given concentrate disposal strategy, also seen in Fig. 4b. The feed-insensitive MVC operation yields increasing 2nd law efficiency values (Fig. S2, Supplementary Information) with increasing feed inlet concentration for a given concentrate disposal or MLD/ZLD scenario, as W_{inp} rises more slowly than the reversible thermodynamic limit at higher freshwater recoveries (Thiel et al., 2015; Tucker et al., 2024). However, it should be noted that in general due to increased non-ideal solution behavior—mainly osmotic pressure and boiling point elevation—and decreased freshwater yield with higher feed salinity, W_{inp} remains highly sensitive to input feed conditions and rated freshwater recovery, with greater specific energy consumption for increased concentrate brine conditions needed for MLD/ZLD. This is evident from the significant W_{inp} increase by about 50–400% from brackish to seawater across all concentrate disposal or MLD and ZLD scenarios. In addition, it can be observed from Fig. 4b and c that the W_{inp} and *WC* are almost an order of magnitude higher for PW feeds, when compared to that for brackish and seawater salinities for a given concentrate handling condition. This clearly highlights the substantial energy and cost requirements for treating hypersaline feeds under MLD and ZLD scenarios, when compared to conventional water treatment applications involving BW and SW.

Similar reasons explain the water treatment cost results in Fig. 4c, which increase with feed inlet concentration from brackish to seawater (30–290%), mainly due to higher osmotic pressure and concentration polarization. This raises membrane area requirements and pushes the optimal $\Delta P_{1,2}$ close to the 65 bar limit (as noted in (Bartholomew et al., 2018)), resulting in higher capital and operating expenses (Table 1) and reduced freshwater yield, causing increased *WC* as per Eq. (11). For PW desalination, cost factors related to W_{inp} and A_{HX} are mathematically independent of *AWP* in the techno-economic model (Bartholomew et al., 2018; Swaminathan et al., 2016). However, factors such as increased compressor pressure ratio from non-linear boiling point elevation and higher brine disposal costs due to increased concentrate and decreased permeate flowrates from Fig. 4a contribute to the slow (compared to brackish and seawater treatment results) but marked *WC* rise for PW feed from 50,000–150,000 ppm. Although this underscores the feed-insensitive benefit of thermal MVC desalination, *WC* was found to asymptotically increase beyond 200,000–230,000 ppm PW feed concentrations for the same reasons in (Katakam et al., n.d.), indicating that sustained energy advantages do not maintain constant *WC* for MVC at higher PW inlet concentrations.

Increasing freshwater recovery to meet MLD and ZLD standards raises water treatment costs for brackish and seawater cases (90% for the former and 470% for the latter from concentrate disposal to ZLD limit desalination scenarios), as shown in Fig. 4c., consistent with SEC trends, due to greater desalination contribution via the costly thermal MVC process handling brine concentrations > 70,000 ppm. A similar non-linear increase in *WC* for MVC-based PW treatment is observed, mainly from higher W_{inp} operational expenses. However, for 91,000 ppm feeds, water treatment costs decrease from MLD lower limit to MLD upper limit concentrate handling scenario before rising again at higher ZLD standards. This reflects a trade-off where increased annual treatment costs are outweighed by almost 2x increase in freshwater recovery (Fig. 4a) for concentrate salinity between 130,000 ppm and 200,000 ppm, causing a *WC* decrease from Eq. (11). A similar trend is obtained for 50,000 ppm when comparing *WC* for conventional disposal and MLD lower limit treatment scenarios. Mainly, the cost benefits vanish at the MLD upper limit and ZLD scenarios, where substantial increase in *AOE* and *ACE* dominate the higher *AWP*, leading to overall higher treatment costs, as also seen in Fig. 4c.

3.2. Comprehensive process-level results for selected desalination case studies

This sub-section details system design and technoeconomic results for two representative water treatment scenarios for data center applications: 1) Hybrid RO-MVC based seawater desalination up to 130,000 ppm concentrate (Fig. 5), and 2) MVC based desalination of mildly hypersaline PW up to 200,000 ppm concentrate MLD limits (Fig. 6). These specific cases are selected for technical elaboration since they represent typical applications for the two conventional and hybrid treatment configurations analyzed presently. Design and operational parameters for cost-optimal two-stage RO-MVC hybrid based on input parameters mentioned in Table 1 are provided in Fig. 5a, while the overall cost distribution is provided in Fig. 5b As it can be observed initial dewatering of seawater feed till 75,000 ppm is achieved from the 2-stage RO process with the first stage operating at much lower ΔP than the 65 bar limit, while the second stage approaches the maximum pressure limit. The two-stage RO benefits from recovery and stage-level pressure redistribution (Section 2.1.1), enabling energy-efficient operation by optimally adjusting pump pressure as required, based on corresponding stage's osmotic pressure profiles, to overcome stage-level osmotic pressure differences and maintain positive water flux. Due to permeate concentration discontinuity, the second stage exhibits higher solute passage (within limits), slightly reducing osmotic pressure difference, increasing flux, lowering membrane area, and reducing overall treatment cost. This cost-optimal two-stage RO has a permeate recovery $RP = 0.53$, with about 325 ppm overall permeate concentration. The permeate is further post-treated based on techniques mentioned in Section 2.4 for increased suitability towards data center end-use. The cost-optimal 2 stage RO has an energy consumption of 1.6 kWh/m^3-permeate and treatment cost of 0.6 \$/m^3, with energy and cost metrics reported normalized with respect to total permeate produced by the hybrid RO-MVC desalination system. Fig. 5b breaks down treatment costs, showing membrane and electricity costs as major contributors due to operation near maximum salinity and second stage operating close to the 65 bar limit. Pretreatment and post-treatment costs contribute up to 30% of the overall water cost, clearly highlighting the non-negligible contribution of those parameters and the importance of including

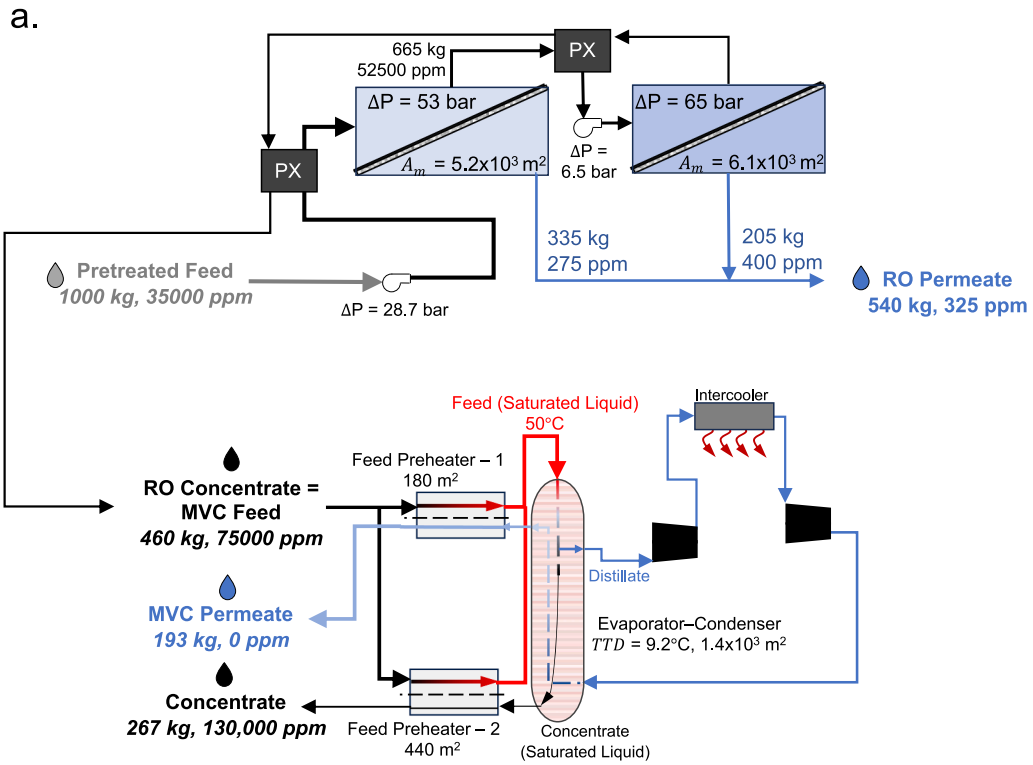


Fig. 5. Results for cost-optimal 2 stage RO-MVC hybrid desalination for MLD-based seawater treatment (Feed TDS= 35,000 ppm). a) Simplified schematic showing system-level design and performance parameters, and b) Pie chart showing breakdown of cost of treatment c) Monte Carlo Sensitivity analysis of water cost for different technical and techno-economic parameters.

them into the techno-economic analysis, as emphasized in the Introduction. The system has a RO process-level second law efficiency of 51.2%, with exergy distribution in Fig. S3 (Supplementary Information), identifying isenthalpic permeate depressurization through the RO membrane as the main irreversibility source (Mistry et al., 2011).

In order to reach MLD standards, the concentrate from the two-stage RO system is further dewatered to 130,000 ppm through MVC desalination operating at $TTD = 9.2$ K as shown in Fig. 5a. This cost-optimal MVC operation results in permeate recovery $RP = 0.42$, 7.8 kWh/m^3-permeate energy consumption, MVC process-level second law efficiency of 8.7% and 2.6$/m^3-permeate treatment cost. The specific energy and treatment cost results for MVC process are based on cumulative permeate production from the overall hybrid desalination process. The thermodynamic state variables for the MVC process are provided in Table S3 of Supplementary Information, with exergy distribution provided in Fig. S3. It should be noted that the exergy distribution values reported in Fig. S3. is based on the normalized freshwater recovery values corresponding to each desalination process and not the entire RO-MVC hybrid. Figure S3 in the Supplementary Information is included to bridge the process-level thermodynamic analysis with the performance of the integrated hybrid RO–MVC system by presenting the overall energy distribution of the hybrid process alongside the exergy distribution of the individual desalination subsystems. The cost distribution of MVC process within the hybrid desalination system is shown in Fig. 5b where in electricity and evaporator costs were the major contributors. Overall, the hybrid two-stage RO-MVC MLD desalination system will provide 736 liters of freshwater suitable for data center end use, for every 1000 kg of seawater being treated, with specific power consumption of 9.4 kWh/m^3-permeate (overall RO-MVC hybrid desalination 2nd law efficiency = 15.9%) and water cost of 3.2 $/m^3-permeate. Due to the energy and cost-intensive operation of MVC treatment processes, as discussed in the previous section and observed in Fig. 5b. and Fig. S4, W_{inp} and WC for the overall RO-MVC hybrid desalination is dominated by that from MVC than initial RO dewatering process.

This also affects the Monte-Carlo sensitivity analysis (±20% variation) shown in Fig. 5c, wherein greater influence of MVC process-related parameter on water cost is observed than that of RO. Energy-cost from electricity (with majority contribution from MVC process itself), followed by MVC evaporator cost and compressor cost are observed to significantly influence the overall water cost-sensitivity of the studied RO-MVC hybrid desalination. These results, along with MVC contribution from Fig. 5b. corroborates the present approach for cost-optimization explained Section 2.1.2, that cost parameters strongly dependent on TTD substantially influence MVC desalination techno-economics. Among other cost parameters, water cost for RO-MVC hybrid was observed to be strongly dependent on discount rate, plant life and MLD brine management cost input for overall RO-MVC techno-economic analysis, followed by RO membrane cost and RO pump cost given the high capital expenditure contribution for membrane technique. With respect to technical parameter influence on water cost, energy-related MVC compressor efficiency followed by RO pump efficiency were observed to drive the overall techno-economics, with MVC compressor parameter sensitivity resulting in an efficiency-cost tradeoff. This is because both lower compressor efficiency, due to increased MVC energy-cost, would drive up the system techno-economics, whereas higher compressor efficiency despite decreased energy demand would also increase the RO-MVC cost due to increased compressor capital expenditure for purchasing higher performance compressor.

Fig. 6a shows a detailed system-level design of the MVC-based PW desalination system, optimized for WC minimization, yielding an optimal TTD= 10 K, as per input parameters in Table 1. Cost-optimal TTD for MVC desalination close to that of the previously discussed hybrid RO-MVC seawater treatment scenario, was also obtained for 50,000 ppm MVC treatment case study. Details regarding thermodynamic state-level variables is provided in Table S.4 of Supplementary Information. A 50,000 ppm PW feed is directly dewatered to the MLD

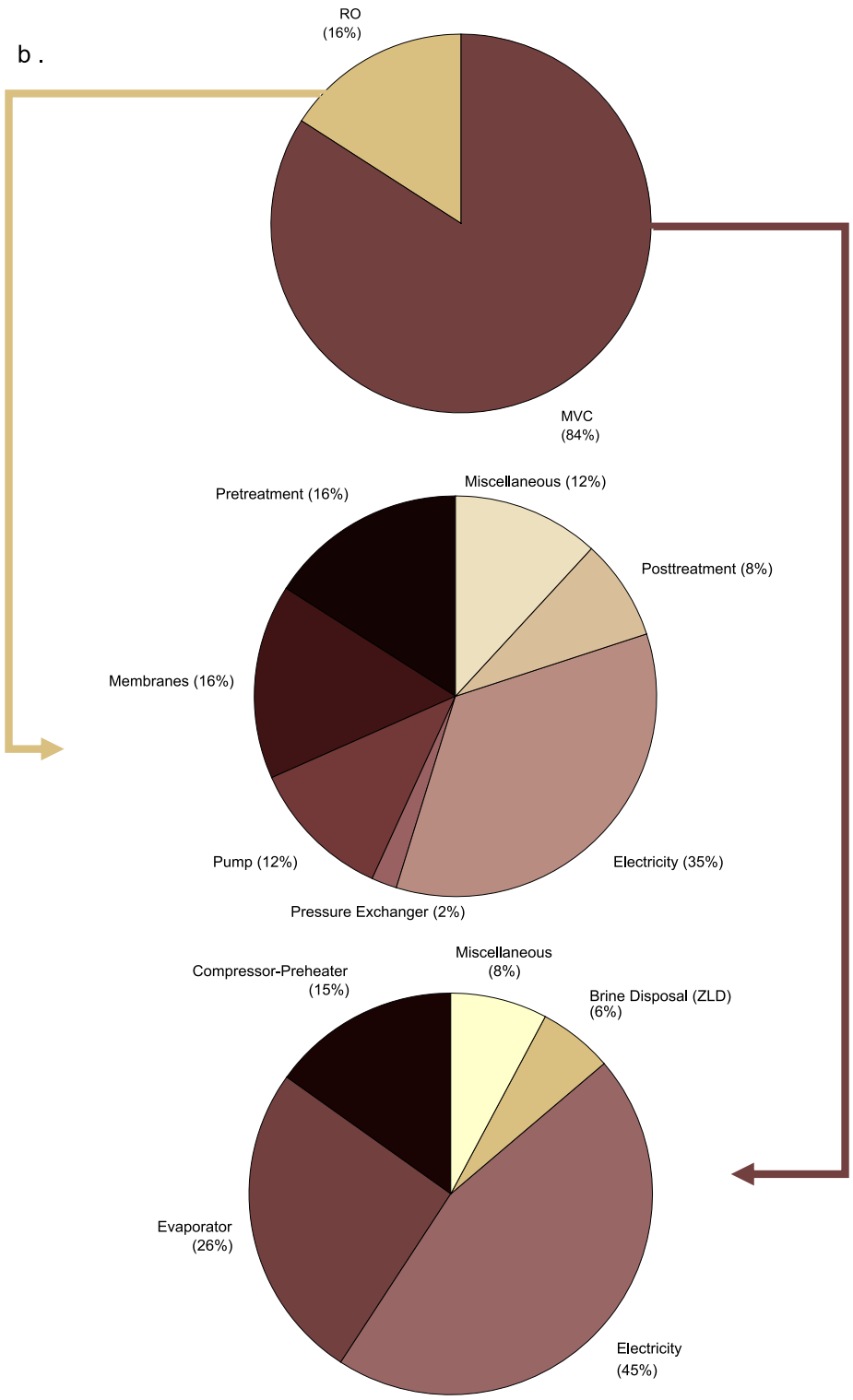


Fig. 5. (*continued*).

upper limit (200,000 ppm) for environmentally benign brine management, producing 754 liters of freshwater suitable for data center applications, with negligible salt concentration, per 1000 kg of PW treated. The system needs W_{inp} of 36 kWh/m^3-permeate, resulting in a low 2nd law efficiency of 6.8% due to energy-intensive phase change separation. This is evident for both the MVC based desalination results obtained in this subsection, wherein from the exergy distribution provided in Fig. S3 and S4 (Supplementary Information), maximum exergy destruction occurs in the evaporator-condenser due to thermal irreversibility for achieving heat transfer with finite temperature difference (Section 2.1.2), followed by mechanical irreversibility in the compressor from friction and viscous effects. Fig. 6b shows a similar distribution of the overall MVC cost of 10.4 \$/m^3-permeate for the cost-optimal design, highlighting that electricity for compressors significantly drives treatment costs due to high W_{inp} and compressor pressure ratios governed by *TTD*. Additionally, evaporator-condenser costs, linked to heat transfer

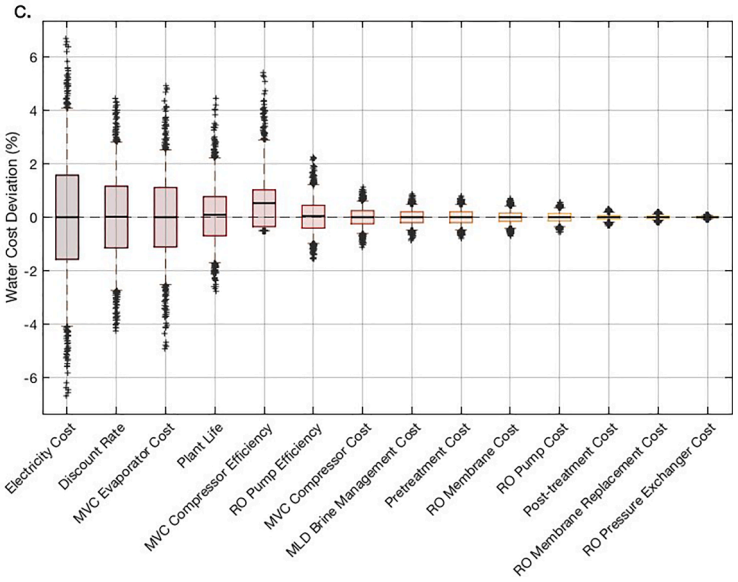


Fig. 5. (*continued*).

parameters and *TTD*, were observed to be the next highest contribution to *WC* (similar to the result obtained for MVC in hybrid desalination scenario), further validating the hypothesis and cost-optimization approach in Section 2.1.2 on the critical role of *TTD* parameter in MVC desalination. Similar corroborating results were observed also from Monte-Carlo sensitivity analysis shown in Fig. 6c for the present MVC desalination system, wherein electricity cost followed by evaporator and compressor cost was observed to strongly influence of the overall technoeconomics. Water cost was also observed to be highly sensitive to discount rate and plant life (similar to the previously discussed RO-MVC hybrid result), followed by compressor efficiency.

3.3. SWUI results for representative emerging data centers

As discussed in Section 2.4, *SWUI* is calculated for representative siting and deployment scenarios for three data centers in i) Mild water stress inland region with brackish water desalination (Case 1), ii) Substantial water stress coastal region with seawater desalination (Case 2), iii) Severely water scarce arid region, with a substantial opportunity of produced water reuse and desalination till ZLD standards (Case 3). Fig. 7a shows the range of *SWUI* values (shades of blue) for these cases through the end of this decade, based on projected *WUE* values (Fig. 3) considering adaptation of thermal management techniques (Shehabi et al., 2024) and pessimistic to optimistic climate change water stress data (Temporal resolution Unimproved /No Drinking Water Analyze Click map Enter Address, 2023). Following the WUI classification (Neudorfer and Llera, 2025), regional water stress impacts based on *SWUI* are categorized by values from 0 to 9 (indicated by background colors in the plot), with values above 9 indicating severe water stress impact from the data center. Fig. 7b quantifies *SWUI* results across all case studies and desalination scenarios for the year 2030, reporting values for high, medium, and low projected *WUE* profiles under pessimistic, business-as-usual, and optimistic *WS* climate conditions. *WUE* values at the medium projected rate are obtained by arithmetic average of high and low *WUE* values for the corresponding year.

SWUI values increase across all case studies through 2030 mainly due to rising *WUE*. However, the *SWUI* band for brackish water desalination in Case 1 only reaches moderate *SWUI* based water stress impact by 2030, owing to higher freshwater recovery ($RR = 931.2$ L/kg-feed) and lower *WS* values ($WS = 1$–3), resulting in the lowest *SWUI* from Eq. (12) among all the cases (Fig. 7a). Conversely, *SWUI* for ZLD-based PW treatment in Case 3 falls in the lower moderate water stress impact *SWUI* range (Fig. 7b), due to increased *WS* (4–5) because of arid geography and lower *RR* (about 653 L/kg-feed) from desalination, both elevating *SWUI*. Conventional seawater treatment for Case 2 raises *SWUI* near the upper moderate *SWUI* impact limit, crossing into substantial water stress impact range by 2029, driven by lower *RR* (482 L/kg-feed) and high *WS* (3–5) of the coastal site.

These results clearly demonstrate the *SWUI* parameter's ability to assess water-related impacts of data centers using a scenario-specific and multifactorial approach, as outlined in Section 2.3. It is evident from Fig. 7a that despite higher *WS* levels and the involvement of hypersaline produced water feed, the overall water stress impact of Case 3. based on *SWUI* is still lower than that of Case 2., which employs conventional seawater desalination. This is due to the lower freshwater recovery achieved in the latter, necessitating increased saline water withdrawals to produce the same quantity of freshwater for data center use, compared to Case 3. Such detailed results, crucial for sustainable siting decisions of the computing infrastructure, would not have been obtained if the effects of saline water characteristics and desalination performance were not integrated into the *SWUI* formulation (Eq. (12)), essentially overlooking the influence of saline water as the actual commodity withdrawn from the nearby environment. For example, evaluating the water stress impact based on the previously defined *WUI* metric for data centers[1] would have suggested Case 3. to be more intensive and environmentally detrimental, compared to Case 2, neglecting the limited freshwater recovery achievable through the conventional seawater desalination over ZLD-based produced water desalination.

An alternative to mitigate the water stress impact of Case 2 could be through increasing freshwater recovery of seawater desalination system till 200,000 ppm MLD, through hybridized coupling with MVC thermal desalination. This will reduce *SWUI* by about 40% due to increased freshwater recovery (Fig. 7b), lowering the *SWUI* values below that of

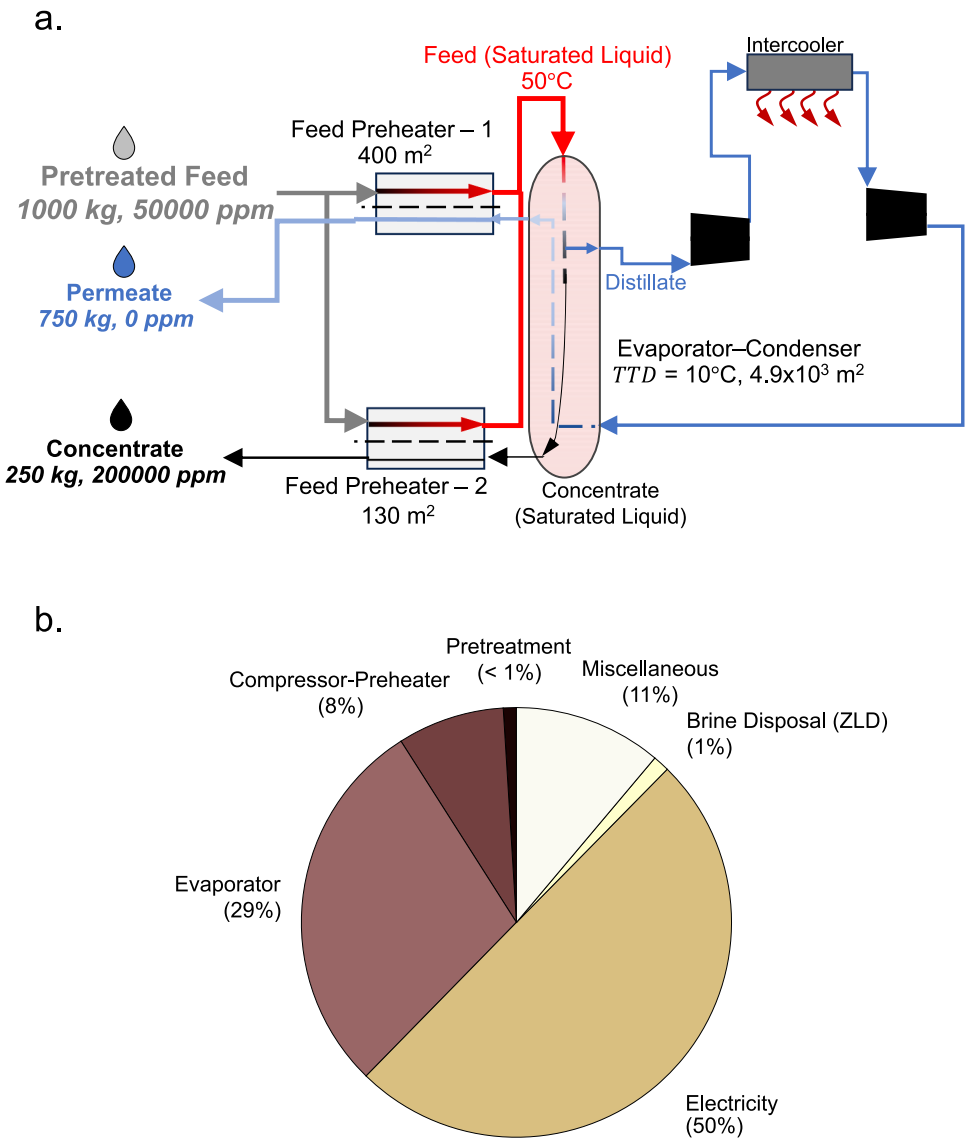


Fig. 6. Results for cost-optimal MVC desalination-based treatment of 50,000 ppm TDS produced water feed till MLD standards. a) Simplified schematic showing system-level design and performance parameters, and b) Pie chart showing breakdown of treatment cost c) Monte Carlo sensitivity analysis of water cost for different technical and techno-economic parameters.

Case 3, thereby resulting in minimal to low range moderate *SWUI*-based water stress impact through the decade. Additionally, sustainable brine management via MLD supports circular resource economy incentives (DuChanois et al., 2023; Nikkhah et al., 2024) instead of hazardous disposal of concentrate through conventional practices. These results highlight the previously un-analyzed tradeoffs with respect to MLD and ZLD desalination systems specific to data center applications. Increased water treatment freshwater recovery for environmentally benign brine management through MLD/ZLD, although would result in increased energy requirements and cost requirements (Section 3.1), it would also result in decreased water-stress impact for the desalination systems. This could especially be useful for data centers located in inland and arid regions with abundant wastewater availability and limited brine disposal options. Desalination energy metrics obtained in Section 3.1 are generally two orders of magnitude lower than the typical power consumption values for data center (Shehabi et al., 2024). This suggests that the increased energy requirements and resulting carbon emissions from achieving MLD/ZLD standards would be negligible compared to those of the end-use application considered. This minimal increase in desalination operating demand would enable effective water-resource utilization for data centers, facilitate viable mineral extraction, and alleviate local hydrological impacts from the computing infrastructure.

It should further be emphasized that incorporating such desalination-specific effects on the overall water footprint implications of data centers cannot be obtained from previous formulations (Neudorfer and Llera, 2025; Azevedo et al., 2011), as it would have resulted in an indifferent result for Case 2 for both conventional disposal or MLD/ZLD based desalination strategy. Additionally, the water stress impact values would have been underpredicted by a factor of $\left(\frac{10^6}{RR\times\rho_{fi}}\right)$, substantially overestimating the sustainable operation of data centers,

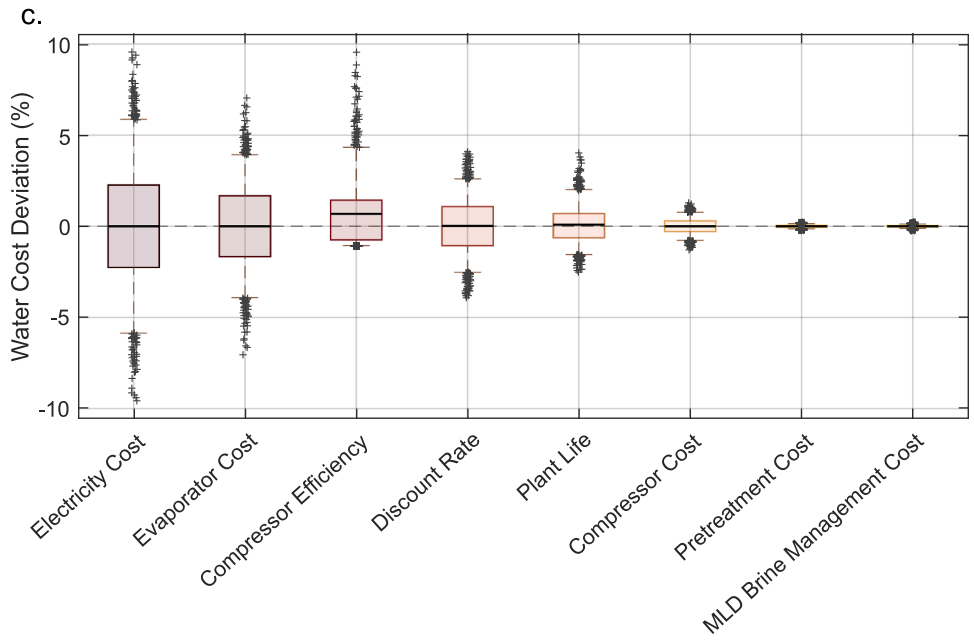


Fig. 6. (*continued*).

especially for scenarios involving high freshwater recoveries (such as Case 1. And Case 3.). Cumulatively, these factors clearly highlight the unified data center water-related modeling capabilities of the proposed *SWUI* analysis framework, which could be utilized as key inputs for an overall data center resource effectiveness performance model (Fukuda et al., 2025), for capturing the related energy, environmental and policy related implications.

It should be noted that for Case 2 (seawater desalination), higher *SWUI* values do not imply that seawater withdrawal depletes freshwater resources. Rather, it reflects that the freshwater produced through desalination is allocated to data center operation within a region where it could alternatively satisfy competing municipal, industrial, or agricultural demands, the significance of which depends on the local water stress. Furthermore, by explicitly incorporating freshwater recovery, present framework inherently incentivizes higher-recovery MLD/ZLD strategies through reduced saline water withdrawal and lower concentrate discharge volumes. For Case 3 (produced water desalination), although reuse of hypersaline oilfield waste is environmentally beneficial, produced water is a by-product of hydraulic fracturing, an inherently water-intensive process that relies on significant regional water withdrawals in already water-stressed inland basins such as the Permian Basin. Consequently, although produced water reuse provides substantial environmental and resource recovery benefits, the oilfield waste is ultimately obtained from a water-intensive process, and the associated data center still operates within a region where freshwater availability remains an important sustainability consideration. However, Case 3 sees substantial benefit, as its higher freshwater recovery directly reduces saline water withdrawal and therefore lowers *SWUI*. The *WS* term contextualizes this benefit within the regional water-stress conditions where the desalination-supported data center is deployed.

A sensitivity analysis on the *SWUI* formulation is provided in Table S2. in the Supplementary Information, comparing mean *SWUI* values between 2023–2030 for Cases 1–3 under linear, quadratic, and exponential formulations of the *WS* parameter, using the projected *WUE* values in Fig. 3. Although an increased overlap between the projected *SWUI* ranges was observed under the nonlinear forms, the overall case-study ranking shown in Fig. 7a and discussed in Section 3.3 is preserved across all three formulations, corroborating the mathematical considerations of the proposed *SWUI* parameter.

Fig. 7 also presents average *SWUI* values for evaporative cooling-based data center thermal management (blue dashed line in Fig. 7a), with values inside bracket in Fig. 7b across all three case studies. For a *WUE* ranging from 1.7–2.5 L/kWh for evaporative cooling, only brackish water desalination in Case 1 maintained *SWUI* below the severe water stress threshold ($SWUI = 7.2$), while all other scenarios exhibited substantially higher *SWUI* values (in the range of 12.9–22.3). This underscores the intensified water stress impact of data center developments using water-intensive thermal management in regions with substantial scarcity or arid/semi-arid regions such as in Case 2 and Case 3. It should be noted that using only *WUE* would have resulted in an indifferent water stress impact result between the three cases, resulting only in a qualitative outcome stating that data centers with evaporative cooling are prone to higher water footprint.

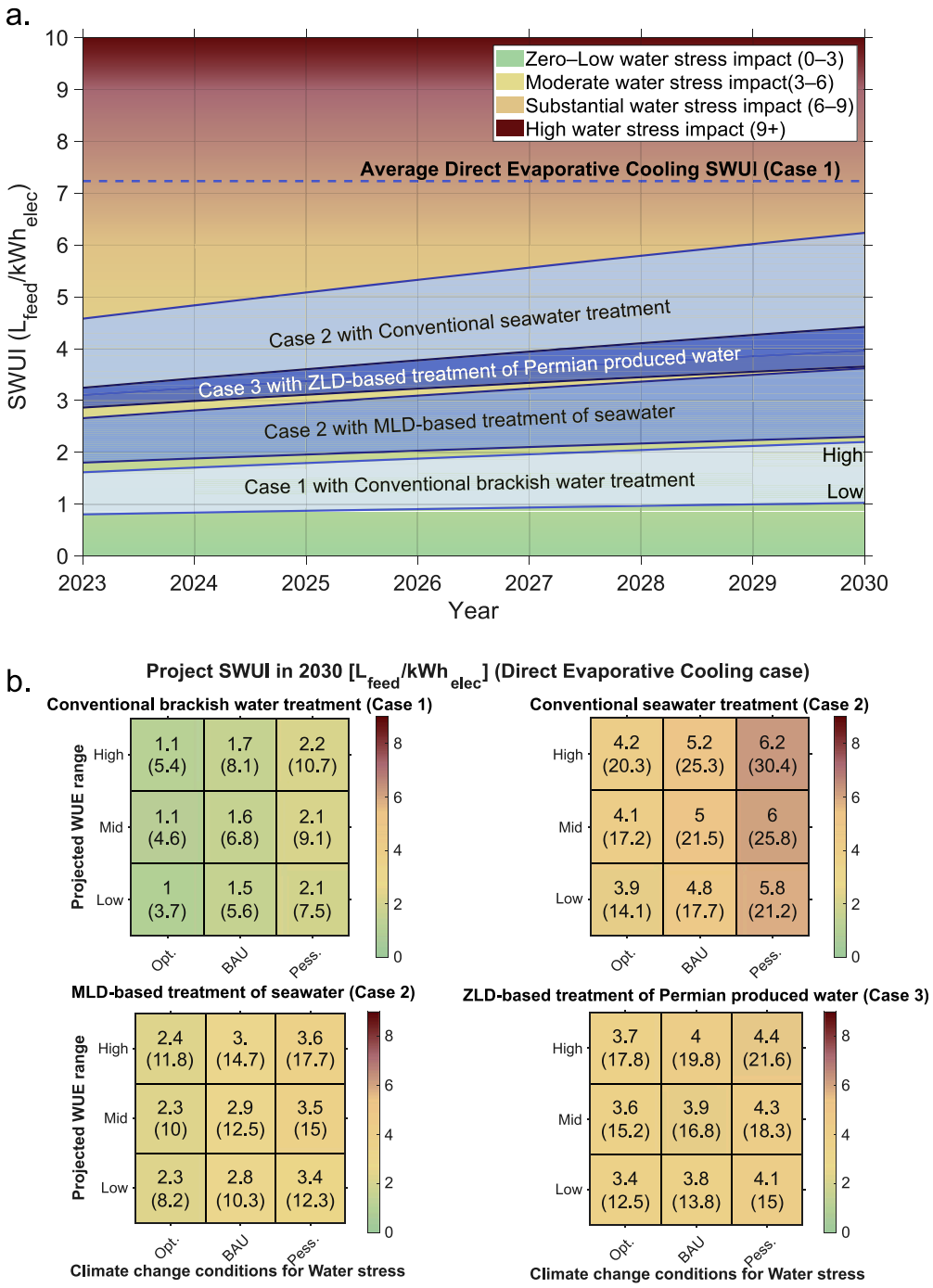


Fig. 7. *SWUI* results for three representative data center deployment case studies: a) Graphical representation of projected *SWUI* from 2023 to 2030 based on averaged *WUE* values across all the thermal management techniques including evaporative cooling and *WUE* values for only evaporative cooling, and b) *SWUI* values calculated for the year 2030 based on different *WUE* projected profiles and Water stress input conditions (References for climate change condition acronyms for *WS* calculation – Opti. = Optimistic; BAU = Business as usual; Pess. = Pessimistic).

4. Conclusions

This study presents system-scale modeling and techno-economic optimization of two-stage reverse osmosis (RO) and Mechanical Vapor Compression (MVC) to produce water for data centers. Treatment of brackish water (BW, 5000 ppm), seawater (SW, 35,000 ppm), and hypersaline produced water (PW, 50,000 and 90,000 ppm) was analyzed with concentrate disposal options being conventional disposal (CD, 70,000 ppm), minimum liquid discharge (MLD, 130,000–200,000 ppm), and zero liquid discharge (ZLD, 200,000–260,000 ppm). For BW and SW, hybrid RO-MVC systems dewatering concentrate to MLD or ZLD increased energy consumption and treatment cost by 53–715% and 44–470%, respectively, compared to conventional RO desalination till CD limits. For PW, MLD/ZLD based MVC-only desalination limited energy consumption and cost increase by <63% and <18%, respectively, when compared to MVC desalination performance operating at conventional disposal limit. Specific case studies include MLD-based hybrid RO-MVC SW desalination (736 L/1000 kg-feed, 9.4 kWh/m^3, 3.2$/m^3, dominated by MVC costs) and MVC desalination of 50,000 ppm PW to MLD (753.7 L/1000 kg-feed, 36 kWh/m^3, 10.4 $/m^3, dominated by

corresponding electricity and evaporator-condenser costs).

A new parameter, Saline Water Utilization Intensity (SWUI) was formulated to integrate data center water utilization effectiveness (WUE) and desalination recovery (RR), along with data center operating conditions to assess the impact of data centers on regional water stress. SWUI was calculated for three representative data centers involving inland to semi-arid/arid hydrological conditions with brackish to hypersaline desalination feed conditions. Projected SWUI increased from 2023 to 2030 due to rising WUE (based on average WUE across all heat rejection and thermal management methods). Additionally, SWUI analysis showed that the water-stress impact of desalination-integrated data centers depends on regional water stress and desalination recovery. By explicitly coupling data-center WUE with saline-water withdrawal, feed salinity, freshwater recovery, and regional water stress, SWUI provides a unified assessment capability that previous WUE/WUI-based frameworks could not capture, particularly for distinguishing conventional disposal from MLD/ZLD desalination pathways. Overall, this study establishes a desalination benchmark linking technical feasibility, cost competitiveness, and regional water-stress implications of brackish-to-hypersaline water treatment for data-centers. The modeling framework supports sustainable data-center deployment and siting decisions, while providing a basis for feasibility comparison and assessment of future developments.

CRediT authorship contribution statement

Vishnu Sree Shanthanu Katakam: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Mariam Arzumanyan:** Writing – review & editing, Resources, Investigation, Conceptualization. **Ning Lin:** Writing – review & editing, Supervision, Software, Resources, Project administration, Funding acquisition. **Vaibhav Bahadur:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.watres.2026.126748.

Data availability

Data will be made available on request.

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